

Qatar-9b

R-2552

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_240104 · created 2026-08-26T01:00:41+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460313.9609 +/- 0.0085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.203 +/- 0.059
Transit depth (Rp/Rs)^2	4.13 +/- 2.41 [%]
Orbital Inclination (inc)	86.16 +/- 1.65
Airmass coefficient 1 (a1)	0.906 +/- 0.072
Airmass coefficient 2 (a2)	0.08 +/- 1.07
Scatter in the residuals of the lightcurve fit is	7.98 %
Best Comparison Star	#2 - [103, 314]
Optimal Aperture	1.53
Optimal Annulus	6.16
Transit Duration (day)	0.0734 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.203 ± 0.059 vs 0.1489 (deviation 0.82σ)
Duration fit vs published	0.0734 d vs 0.0779 d (deviation 0.5σ)
Mid-time O–C	-8.0 min
Depth SNR	3.1
Residual scatter	7.98 %

Light curve

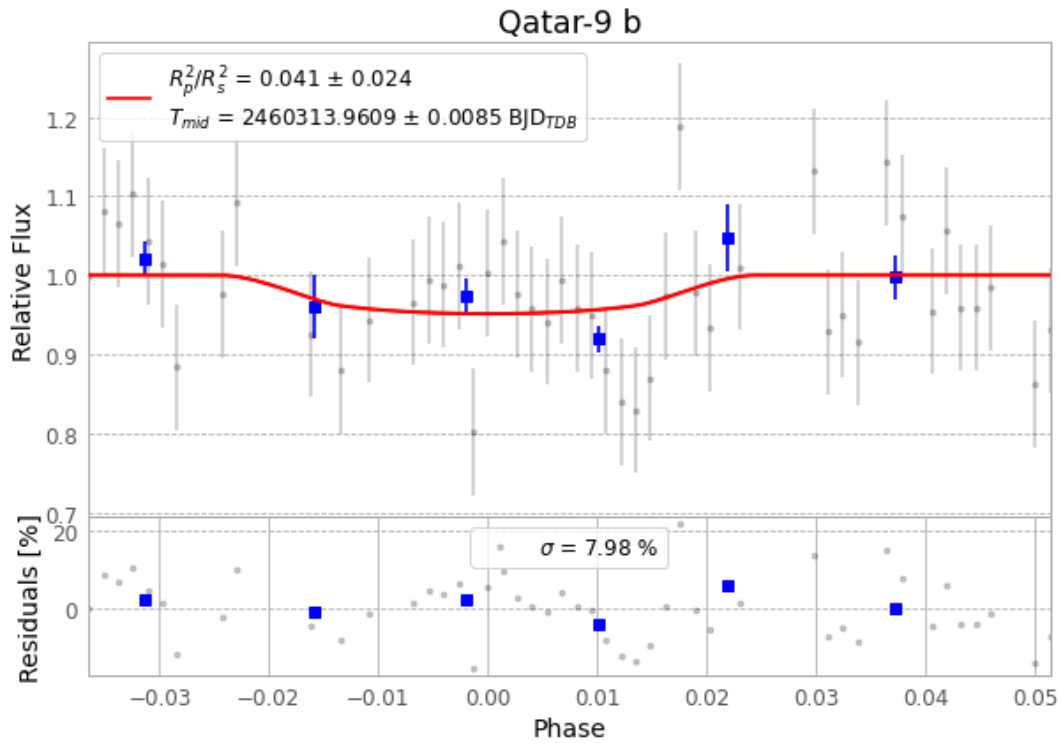


Figure 1 — FinalLightCurve_Qatar-9 b_2024-01-04.png (SHA-256 9baa243cd6fb83ab...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 58856194482dee3a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

68 FITS frames (45 MB), Qatar-9240104093615.FITS → Qatar-9240104125715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-240104.log (196f59c13941...), FinalLightCurve_Qatar-9 b_2024-01-04.png (9baa243cd6fb...), FinalLightCurve_Qatar-9 b_2024-01-04.csv (2b1a42ff1ab0...)

Compute resources

Wall time 128.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 01:00Z.